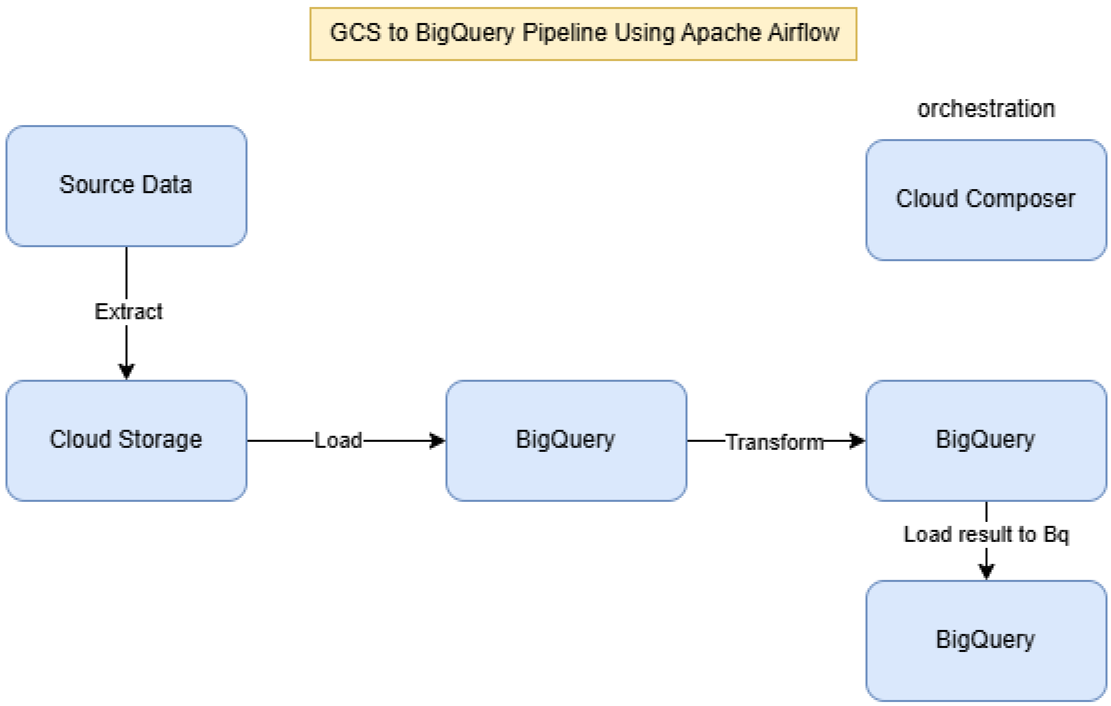
GCS to BigQuery Pipeline Using Apache Airflow

Overview

This document describes the architecture and workflow of a GCS to BigQuery data pipeline orchestrated using Apache Airflow. The pipeline is designed to automate the process of extracting, loading, and transforming (ELT) data from Google Cloud Storage (GCS) into BigQuery for analytics and reporting.



Pipeline Architecture

The pipeline follows a structured ELT (Extract, Load, Transform) process, as illustrated in the architecture diagram. The key components involved are:

1. Source Data
    - Raw data is uploaded to Google Cloud Storage (GCS) in formats such as CSV, JSON, or Parquet.
    - The data can come from various sources, including application logs, transactional databases, or third-party APIs.
2. Cloud Storage (GCS)
    - Acts as a staging area for incoming raw data before processing.
    - Airflow monitors this storage bucket for new files and triggers the data pipeline.

3. Apache Airflow (Cloud Composer) – Orchestration
    - Cloud Composer, a managed Airflow service, orchestrates the data pipeline.
    - It defines a Directed Acyclic Graph (DAG) to automate the extraction, loading, and transformation steps.
4. BigQuery (Loading Phase)
    - The raw data is loaded from GCS into a staging table in BigQuery.
    - The process uses Apache Airflow's GCSToBigQueryOperator to ensure efficient data ingestion.
5. BigQuery (Transformation Phase)
    - Data transformation and cleansing occur within BigQuery using SQL-based transformations.
    - Common transformations include:
        - Data type conversion
        - Aggregation and deduplication
        - Schema validation and enrichment
6. BigQuery (Final Storage & Reporting)
    - The transformed data is stored in the final BigQuery tables.
    - The data is then available for business intelligence (BI) tools such as:
        - Looker
        - Google Data Studio
        - Power BI
    - These tools fetch processed data from BigQuery to generate dashboards and reports.

Workflow Breakdown

1. Extract
    - Airflow detects new files in GCS and initiates the pipeline.
    - The data is read and prepared for loading.
2. Load
    - The extracted data is loaded into BigQuery staging tables.
    - The process can be configured for batch or streaming ingestion.
3. Transform

    - SQL-based transformations run within BigQuery to clean and structure the data.
    - The final data is inserted into a BigQuery reporting table.
4. Data Consumption & Visualization
    - Business teams and analysts can query BigQuery tables directly.
    - The data is visualized using dashboards for insights and decision-making.

Benefits of the Pipeline

- Scalability: Supports large datasets with efficient parallel processing.
- Automation: Fully orchestrated using Apache Airflow (Cloud Composer).
- Cost-Effective: Utilizes serverless BigQuery for transformation and storage.
- Real-time & Batch Processing: Can handle both batch and streaming data.

Conclusion

This GCS to BigQuery pipeline using Apache Airflow enables seamless data processing and analytics. By automating the ELT workflow, businesses can ensure timely insights, improved data governance, and efficient reporting.